Drashti Bhavsar

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Software & ML Engineer · Available full time Jan–Aug 2027 for an internship or co-op · Open to relocation

Education

Northeastern University, Boston, MA — *MS in Computer Science* Aug 2025 – Dec 2027 (expected)

Pandit Deendayal Energy University, India — *B.Tech in Computer Engineering* May 2024
Coursework: Data Structures & Algorithms, Distributed Systems, OOP, Machine Learning, Deep Learning, NLP

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, R
Backend & Systems: FastAPI, Django, Node.js, React, REST, GraphQL, WebSockets, Microservices, Kafka, Redis
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker), GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
ML & GenAI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LangChain, LangGraph, RAG, OpenCV, Deep RL, MLflow
Data: PostgreSQL, MongoDB, Elasticsearch, Milvus, Qdrant, pgvector, PySpark, Airflow, Power BI

Experience

Northeastern University — *Research Assistant, AI/ML & Health Misinformation* Boston, MA · Jan 2026 – May 2026
- Co-built a **28-model framework** (RoBERTa sentiment/emotion, UMAP/HDBSCAN clustering, NLI zero-shot, Llama 3, Mistral, BioMistral, GatorTron) to detect health misinformation in **9,838 Reddit posts**; published at **ICWSM 2026 (AAAI)**.

LOTI AI Inc. — *Machine Learning Engineer Intern* Seattle, WA · Aug 2024 – Jul 2025
- Engineered a production Python inference platform with rate limiting and monitoring: **99% uptime, 10K+ daily requests**.
- Architected Docker/Kubernetes inference microservices with CI/CD, cutting deploys from **2 hours to 15 minutes**.
- Built a multi-scale PyTorch content classifier at **87.3% accuracy, 12% above the production baseline**.
- Built semantic retrieval and ranking for AI prompt routing (BGE, Qwen, SPLADE, ColBERT on Milvus) at **97% recall**.
- Built a real-time EfficientNet + Molmo deepfake detector; kept **95%+ test coverage** on a 5-person Agile team.

Indian School of Business — *Data Analyst* India · Feb 2025 – Jul 2025
- Classified **10,000+ patents** by AI relevance and gender representation with BERT and spaCy NER at **92% accuracy**.
- Automated data collection and ETL, cutting manual effort by **50%**; built dashboards on **20+ years** of trends.

IIT Roorkee, Google ExploreCSR — *Junior Research Fellow (top 50 selected)* India · Jan 2024 – Jun 2024
- Built hierarchical deep RL agents in CARLA for overtaking: **94% success, <50 ms inference, +25% accuracy**.

Virtual Reality Systems — *Software Engineer Intern* India · Feb 2024 – Apr 2024
- Built a React + FastAPI/Django app with REST APIs for **500+ daily users**, cutting API latency **800 → 200 ms**.
- Automated testing and deploys with GitHub Actions and Docker; built a Power BI dashboard (**+30%** efficiency).

ISRO, Space Applications Centre — *Data Science Research Intern* India · Jun 2023 – Aug 2023
- Predicted thunderstorm signal anomalies from NavIC S- and L-band data with Random Forest and LSTM (**98% / 97%**).
- Deployed a real-time L5-signal classifier with FastAPI, improving positioning accuracy by **23%**.

Projects

Real-Time Collaborative Code Editor | React, Node.js, WebSockets, Redis, PostgreSQL, AWS EC2
- **50+ concurrent users at ~100 ms** via operational transformation (99.9% consistency); Redis cut DB load **60%**.

FIFA 2026 World Cup Predictor | XGBoost, Random Forest, FastAPI, Next.js
- Ensemble on **35,304 matches, 97 features** (0.826 log loss, 0.018 ECE) powering **10,000 Monte Carlo** tournament runs.

Patent Claim Hybrid Search Engine | Python, BM25, Embeddings, System Design
- Hybrid BM25 + semantic search with metadata filters, plus a recall and re-rank design that scales to **10M patents**.

Publications & Leadership · Google Scholar

- **5 peer-reviewed papers, 164 citations (h-index 4):** Health Misinformation Detection Using Multi-Model LLM Ensembles (*ICWSM 2026 Workshops, AAAI*); Smart Drip & Sprinkler Irrigation Systems (*Elsevier*, 111 citations); AI-Enabled Dental Caries Detection with Grad-CAM (*Springer*); Milk Quality Prediction Using ML (*EAI*); Industry 4.0 in Formula 1 (*Springer*).
- **Head of Events, ACM at Northeastern** (2026–present) · Google ExploreCSR top-50 fellow · Academic Merit & Research Scholarship · ACM India Summer & Winter Schools (IIT Gandhinagar, IIT Patna).